

Pathology-Level ML Model Performance Summary (Test Set)
Classical ML models trained on neural network pathology predictions

Neural Network Model	Pathology	ML Algorithm	AUC Mean	AUC Std	AUC 95% CI	F1 Mean	F1 Std	Optimal Threshold	Threshold Std	Samples	Positive Rate	Cross-Val Folds
Attention Pool Extra3 Full T	A-lines	Random Forest	0.755	0.021	[0.728-0.781]	0.666	0.022	0.486	0.009	43689	0.472	5
Attention Pool Extra3 Full T	A-lines	Logistic Regression	0.793	0.013	[0.777-0.809]	0.703	0.033	0.532	0.013	43689	0.472	5
Attention Pool Extra3 Full T	A-lines	XGBoost	0.799	0.014	[0.781-0.817]	0.707	0.021	0.472	0.018	43689	0.472	5
Attention Pool Extra3 Full T	Large Consolidation	Random Forest	0.861	0.021	[0.835-0.887]	0.514	0.084	0.398	0.016	43689	0.110	5
Attention Pool Extra3 Full T	Large Consolidation	Logistic Regression	0.920	0.005	[0.913-0.926]	0.519	0.044	0.388	0.015	43689	0.110	5
Attention Pool Extra3 Full T	Large Consolidation	XGBoost	0.904	0.012	[0.890-0.919]	0.561	0.046	0.142	0.008	43689	0.110	5
Attention Pool Extra3 Full T	Pleural Effusion	Random Forest	0.669	0.038	[0.623-0.716]	0.146	0.031	0.254	0.018	43689	0.011	5
Attention Pool Extra3 Full T	Pleural Effusion	Logistic Regression	0.810	0.041	[0.759-0.861]	0.059	0.040	0.470	0.027	43689	0.011	5
Attention Pool Extra3 Full T	Pleural Effusion	XGBoost	0.834	0.040	[0.785-0.883]	0.122	0.040	0.050	0.012	43689	0.011	5
Attention Pool Extra3 Full T	Emphysema and Small Consolidation	Random Forest	0.644	0.013	[0.627-0.660]	0.531	0.038	0.444	0.011	43689	0.405	5
Attention Pool Extra3 Full T	Emphysema and Small Consolidation	Logistic Regression	0.685	0.022	[0.658-0.713]	0.588	0.044	0.492	0.013	43689	0.405	5
Attention Pool Extra3 Full T	Emphysema and Small Consolidation	XGBoost	0.686	0.016	[0.665-0.706]	0.569	0.047	0.438	0.008	43689	0.405	5